

Event	M ($M_\odot$)	$\mathcal{M}$ ($M_\odot$)	m_1 ($M_\odot$)	m_2 ($M_\odot$)	χ_{eff}	D_L (Gpc)	z	M_f ($M_\odot$)	χ_f	$\Delta\Omega$ (deg 2)	SNR
GW190408-181802	$43.0^{+4.2}_{-3.0}$	$18.3^{+1.9}_{-1.2}$	$24.6^{+5.1}_{-3.4}$	$18.4^{+3.3}_{-3.6}$	$-0.03^{+0.14}_{-0.19}$	$1.55^{+0.40}_{-0.60}$	$0.29^{+0.06}_{-0.10}$	$41.1^{+3.9}_{-2.8}$	$0.67^{+0.06}_{-0.07}$	150	$15.3^{+0.2}_{-0.3}$
GW190412	$38.4^{+3.8}_{-3.7}$	$13.3^{+0.4}_{-0.3}$	$30.1^{+4.7}_{-5.1}$	$8.3^{+1.6}_{-0.9}$	$0.25^{+0.08}_{-0.11}$	$0.74^{+0.14}_{-0.17}$	$0.15^{+0.03}_{-0.03}$	$37.3^{+3.9}_{-3.8}$	$0.67^{+0.05}_{-0.06}$	21	$18.9^{+0.2}_{-0.3}$
GW190413-052954	$58.6^{+13.3}_{-9.7}$	$24.6^{+5.5}_{-4.1}$	$34.7^{+12.6}_{-8.1}$	$23.7^{+7.3}_{-6.7}$	$-0.01^{+0.29}_{-0.34}$	$3.55^{+2.27}_{-1.66}$	$0.59^{+0.29}_{-0.24}$	$56.0^{+12.5}_{-9.2}$	$0.68^{+0.12}_{-0.13}$	1500	$8.9^{+0.4}_{-0.7}$
GW190413-134308	$78.8^{+17.4}_{-11.9}$	$33.0^{+8.2}_{-5.4}$	$47.5^{+13.5}_{-10.7}$	$31.8^{+11.7}_{-10.8}$	$-0.03^{+0.25}_{-0.29}$	$4.45^{+2.48}_{-2.12}$	$0.71^{+0.31}_{-0.30}$	$75.5^{+16.4}_{-11.4}$	$0.68^{+0.10}_{-0.12}$	730	$10.0^{+0.4}_{-0.5}$
GW190421-213856	$72.9^{+13.4}_{-9.2}$	$31.2^{+5.9}_{-4.2}$	$41.3^{+10.4}_{-6.9}$	$31.9^{+8.0}_{-8.8}$	$-0.06^{+0.22}_{-0.27}$	$2.88^{+1.37}_{-1.38}$	$0.49^{+0.19}_{-0.21}$	$69.7^{+12.5}_{-8.7}$	$0.67^{+0.10}_{-0.11}$	1200	$10.7^{+0.2}_{-0.4}$
GW190424-180648	$72.6^{+13.3}_{-10.7}$	$31.0^{+5.8}_{-4.6}$	$40.5^{+11.1}_{-7.3}$	$31.8^{+7.6}_{-7.7}$	$0.13^{+0.22}_{-0.22}$	$2.20^{+1.58}_{-1.16}$	$0.39^{+0.23}_{-0.19}$	$68.9^{+12.4}_{-10.1}$	$0.74^{+0.09}_{-0.09}$	28000	$10.4^{+0.2}_{-0.4}$
GW190425	$3.4^{+0.3}_{-0.1}$	$1.44^{+0.02}_{-0.02}$	$2.0^{+0.6}_{-0.3}$	$1.4^{+0.3}_{-0.3}$	$0.06^{+0.11}_{-0.05}$	$0.16^{+0.07}_{-0.07}$	$0.03^{+0.01}_{-0.02}$	—	—	10000	$12.4^{+0.3}_{-0.4}$
GW190426-152155	$7.2^{+3.5}_{-1.5}$	$2.41^{+0.08}_{-0.08}$	$5.7^{+3.9}_{-2.3}$	$1.5^{+0.8}_{-0.5}$	$-0.03^{+0.32}_{-0.30}$	$0.37^{+0.18}_{-0.16}$	$0.08^{+0.04}_{-0.03}$	—	—	1300	$8.7^{+0.5}_{-0.6}$
GW190503-185404	$71.7^{+9.4}_{-8.3}$	$30.2^{+4.2}_{-4.2}$	$43.3^{+9.2}_{-8.0}$	$28.4^{+7.7}_{-8.0}$	$-0.03^{+0.20}_{-0.26}$	$1.45^{+0.69}_{-0.63}$	$0.27^{+0.11}_{-0.11}$	$68.6^{+8.8}_{-7.7}$	$0.66^{+0.09}_{-0.12}$	94	$12.4^{+0.2}_{-0.3}$
GW190512-180714	$35.9^{+3.8}_{-3.5}$	$14.6^{+1.3}_{-1.0}$	$23.3^{+5.3}_{-5.8}$	$12.6^{+3.6}_{-2.5}$	$0.03^{+0.12}_{-0.13}$	$1.43^{+0.55}_{-0.55}$	$0.27^{+0.09}_{-0.10}$	$34.5^{+3.8}_{-3.5}$	$0.65^{+0.07}_{-0.07}$	220	$12.2^{+0.2}_{-0.4}$
GW190513-205428	$53.9^{+8.6}_{-5.9}$	$21.6^{+3.8}_{-1.9}$	$35.7^{+9.2}_{-9.2}$	$18.0^{+7.7}_{-4.1}$	$0.11^{+0.28}_{-0.17}$	$2.06^{+0.88}_{-0.80}$	$0.37^{+0.13}_{-0.13}$	$51.6^{+8.2}_{-5.8}$	$0.68^{+0.14}_{-0.12}$	520	$12.9^{+0.3}_{-0.4}$
GW190514-065416	$67.2^{+18.7}_{-10.8}$	$28.5^{+7.9}_{-4.8}$	$39.0^{+14.7}_{-8.2}$	$28.4^{+9.3}_{-8.8}$	$-0.19^{+0.29}_{-0.32}$	$4.13^{+2.65}_{-2.17}$	$0.67^{+0.33}_{-0.31}$	$64.5^{+17.9}_{-10.4}$	$0.63^{+0.11}_{-0.15}$	3000	$8.2^{+0.3}_{-0.6}$
GW190517-055101	$63.5^{+9.6}_{-9.6}$	$26.6^{+4.0}_{-4.0}$	$37.4^{+11.7}_{-7.6}$	$25.3^{+7.0}_{-7.3}$	$0.52^{+0.19}_{-0.19}$	$1.86^{+1.62}_{-0.84}$	$0.34^{+0.24}_{-0.14}$	$59.3^{+9.1}_{-8.9}$	$0.87^{+0.05}_{-0.07}$	470	$10.7^{+0.4}_{-0.6}$
GW190519-153544	$106.6^{+13.5}_{-14.8}$	$44.5^{+6.4}_{-7.1}$	$66.0^{+10.7}_{-12.0}$	$40.5^{+11.0}_{-11.1}$	$0.31^{+0.20}_{-0.22}$	$2.53^{+1.83}_{-0.92}$	$0.44^{+0.25}_{-0.14}$	$101.0^{+12.4}_{-13.8}$	$0.79^{+0.07}_{-0.13}$	860	$15.6^{+0.2}_{-0.3}$
GW190521	$163.9^{+39.2}_{-23.5}$	$69.2^{+17.0}_{-10.6}$	$95.3^{+28.7}_{-18.9}$	$69.0^{+22.7}_{-23.1}$	$0.03^{+0.32}_{-0.39}$	$3.92^{+2.19}_{-1.95}$	$0.64^{+0.28}_{-0.28}$	$156.3^{+36.8}_{-22.4}$	$0.71^{+0.12}_{-0.16}$	1000	$14.2^{+0.3}_{-0.3}$
GW190521-074359	$74.7^{+7.0}_{-4.8}$	$32.1^{+3.2}_{-2.5}$	$42.2^{+5.9}_{-4.8}$	$32.8^{+5.4}_{-6.4}$	$0.09^{+0.10}_{-0.13}$	$1.24^{+0.40}_{-0.57}$	$0.24^{+0.07}_{-0.10}$	$71.0^{+6.5}_{-4.4}$	$0.72^{+0.05}_{-0.07}$	550	$25.8^{+0.1}_{-0.2}$
GW190527-092055	$59.1^{+21.3}_{-9.8}$	$24.3^{+9.1}_{-4.2}$	$36.5^{+16.4}_{-9.0}$	$22.6^{+10.5}_{-8.1}$	$0.11^{+0.28}_{-0.28}$	$2.49^{+2.48}_{-1.24}$	$0.44^{+0.34}_{-0.20}$	$56.4^{+20.2}_{-9.3}$	$0.71^{+0.12}_{-0.16}$	3700	$8.1^{+0.3}_{-0.9}$
GW190602-175927	$116.3^{+19.0}_{-15.6}$	$49.1^{+9.1}_{-8.5}$	$69.1^{+15.7}_{-13.0}$	$47.8^{+14.3}_{-17.4}$	$0.07^{+0.25}_{-0.24}$	$2.69^{+1.79}_{-1.12}$	$0.47^{+0.25}_{-0.17}$	$110.9^{+17.7}_{-14.9}$	$0.70^{+0.10}_{-0.14}$	690	$12.8^{+0.2}_{-0.3}$
GW190620-030421	$92.1^{+18.5}_{-13.1}$	$38.3^{+8.3}_{-6.5}$	$57.1^{+16.0}_{-12.7}$	$35.5^{+12.2}_{-12.3}$	$0.33^{+0.22}_{-0.25}$	$2.81^{+1.68}_{-1.31}$	$0.49^{+0.23}_{-0.20}$	$87.2^{+16.8}_{-12.1}$	$0.79^{+0.08}_{-0.15}$	7200	$12.1^{+0.3}_{-0.4}$
GW190630-185205	$59.1^{+4.6}_{-4.8}$	$24.9^{+2.1}_{-2.1}$	$35.1^{+6.9}_{-5.6}$	$23.7^{+5.2}_{-5.1}$	$0.10^{+0.12}_{-0.13}$	$0.89^{+0.56}_{-0.37}$	$0.18^{+0.10}_{-0.07}$	$56.4^{+4.4}_{-4.6}$	$0.70^{+0.05}_{-0.07}$	1200	$15.6^{+0.2}_{-0.3}$
GW190701-203306	$94.3^{+12.1}_{-9.5}$	$40.3^{+5.4}_{-4.9}$	$53.9^{+11.8}_{-8.0}$	$40.8^{+8.7}_{-12.0}$	$-0.07^{+0.23}_{-0.29}$	$2.06^{+0.76}_{-0.73}$	$0.37^{+0.11}_{-0.12}$	$90.2^{+11.3}_{-8.9}$	$0.66^{+0.09}_{-0.13}$	46	$11.3^{+0.2}_{-0.3}$
GW190706-222641	$104.1^{+20.2}_{-13.9}$	$42.7^{+10.0}_{-7.0}$	$67.0^{+14.6}_{-16.2}$	$38.2^{+14.6}_{-13.3}$	$0.28^{+0.26}_{-0.29}$	$4.42^{+2.59}_{-1.93}$	$0.71^{+0.32}_{-0.27}$	$99.0^{+18.3}_{-13.5}$	$0.78^{+0.09}_{-0.18}$	650	$12.6^{+0.2}_{-0.4}$
GW190707-093326	$20.1^{+1.9}_{-1.3}$	$8.5^{+0.6}_{-0.5}$	$11.6^{+3.3}_{-1.7}$	$8.4^{+1.4}_{-1.7}$	$-0.05^{+0.10}_{-0.08}$	$0.77^{+0.38}_{-0.37}$	$0.16^{+0.07}_{-0.07}$	$19.2^{+1.9}_{-1.3}$	$0.66^{+0.03}_{-0.04}$	1300	$13.3^{+0.2}_{-0.4}$
GW190708-232457	$30.9^{+2.5}_{-1.8}$	$13.2^{+0.9}_{-0.6}$	$17.6^{+4.7}_{-2.3}$	$13.2^{+2.0}_{-2.7}$	$0.02^{+0.10}_{-0.08}$	$0.88^{+0.33}_{-0.39}$	$0.18^{+0.06}_{-0.07}$	$29.5^{+2.5}_{-1.8}$	$0.69^{+0.04}_{-0.04}$	14000	$13.1^{+0.2}_{-0.3}$
GW190719-215514	$57.8^{+18.3}_{-10.7}$	$23.5^{+6.5}_{-4.0}$	$36.5^{+18.0}_{-10.3}$	$20.8^{+9.0}_{-7.2}$	$0.32^{+0.29}_{-0.31}$	$3.94^{+2.59}_{-2.00}$	$0.64^{+0.33}_{-0.29}$	$54.9^{+17.3}_{-10.2}$	$0.78^{+0.11}_{-0.17}$	2900	$8.3^{+0.3}_{-0.8}$
GW190720-000836	$21.5^{+4.3}_{-2.3}$	$8.9^{+0.5}_{-0.8}$	$13.4^{+6.7}_{-3.0}$	$7.8^{+2.3}_{-2.2}$	$0.18^{+0.14}_{-0.12}$	$0.79^{+0.69}_{-0.32}$	$0.16^{+0.12}_{-0.06}$	$20.4^{+4.5}_{-2.2}$	$0.72^{+0.06}_{-0.05}$	460	$11.0^{+0.3}_{-0.7}$
GW190727-060333	$67.1^{+11.7}_{-8.0}$	$28.6^{+5.3}_{-3.7}$	$38.0^{+9.5}_{-6.2}$	$29.4^{+7.1}_{-8.4}$	$0.11^{+0.26}_{-0.25}$	$3.30^{+1.54}_{-1.50}$	$0.55^{+0.21}_{-0.22}$	$63.8^{+10.9}_{-7.5}$	$0.73^{+0.10}_{-0.10}$	830	$11.9^{+0.3}_{-0.5}$
GW190728-064510	$20.6^{+4.5}_{-1.3}$	$8.6^{+0.5}_{-0.3}$	$12.3^{+7.2}_{-2.2}$	$8.1^{+1.7}_{-2.6}$	$0.12^{+0.20}_{-0.07}$	$0.87^{+0.26}_{-0.37}$	$0.18^{+0.05}_{-0.07}$	$19.6^{+4.7}_{-1.3}$	$0.71^{+0.04}_{-0.04}$	400	$13.0^{+0.2}_{-0.4}$
GW190731-140936	$70.1^{+15.8}_{-11.3}$	$29.5^{+7.1}_{-5.2}$	$41.5^{+12.2}_{-9.0}$	$28.8^{+9.7}_{-9.5}$	$0.06^{+0.24}_{-0.24}$	$3.30^{+2.39}_{-1.72}$	$0.55^{+0.31}_{-0.26}$	$67.0^{+14.6}_{-10.8}$	$0.70^{+0.10}_{-0.13}$	3400	$8.7^{+0.2}_{-0.5}$
GW190803-022701	$64.5^{+12.6}_{-9.0}$	$27.3^{+5.7}_{-4.1}$	$37.3^{+10.6}_{-7.0}$	$27.3^{+7.8}_{-8.2}$	$-0.03^{+0.24}_{-0.27}$	$3.27^{+1.95}_{-1.58}$	$0.55^{+0.26}_{-0.24}$	$61.7^{+11.8}_{-8.5}$	$0.68^{+0.10}_{-0.11}$	1500	$8.6^{+0.3}_{-0.5}$
GW190814	$25.8^{+1.0}_{-0.9}$	$6.09^{+0.06}_{-0.06}$	$23.2^{+1.1}_{-1.0}$	$2.59^{+0.08}_{-0.09}$	$0.00^{+0.06}_{-0.06}$	$0.24^{+0.04}_{-0.05}$	$0.05^{+0.009}_{-0.010}$	$25.6^{+1.1}_{-0.9}$	$0.28^{+0.02}_{-0.02}$	19	$24.9^{+0.1}_{-0.2}$
GW190828-063405	$58.0^{+7.7}_{-4.8}$	$25.0^{+3.4}_{-2.1}$	$32.1^{+5.8}_{-4.0}$	$26.2^{+4.6}_{-4.8}$	$0.19^{+0.15}_{-0.16}$	$2.13^{+0.66}_{-0.93}$	$0.38^{+0.10}_{-0.15}$	$54.9^{+7.2}_{-4.3}$	$0.75^{+0.06}_{-0.07}$	520	$16.2^{+0.2}_{-0.3}$
GW190828-065509	$34.4^{+5.4}_{-4.4}$	$13.3^{+1.2}_{-1.0}$	$24.1^{+7.0}_{-7.2}$	$10.2^{+3.6}_{-2.1}$	$0.08^{+0.16}_{-0.16}$	$1.60^{+0.62}_{-0.60}$	$0.30^{+0.10}_{-0.10}$	$33.1^{+5.5}_{-4.5}$	$0.65^{+0.08}_{-0.08}$	660	$10.0^{+0.3}_{-0.5}$
GW190909-114149	$75.0^{+55.9}_{-17.6}$	$30.9^{+17.2}_{-7.5}$	$45.8^{+52.7}_{-13.3}$	$28.3^{+13.4}_{-12.7}$	$-0.06^{+0.37}_{-0.36}$	$3.77^{+3.27}_{-2.22}$	$0.62^{+0.41}_{-0.33}$	$72.0^{+54.9}_{-16.8}$	$0.66^{+0.15}_{-0.20}$	4700	$8.1^{+0.4}_{-0.6}$
GW190910-112807	$79.6^{+9.3}_{-9.1}$	$34.3^{+4.1}_{-4.1}$	$43.9^{+7.6}_{-6.1}$	$35.6^{+6.3}_{-7.2}$	$0.02^{+0.18}_{-0.18}$	$1.46^{+1.03}_{-0.58}$	$0.28^{+0.16}_{-0.10}$	$75.8^{+8.5}_{-8.6}$	$0.70^{+0.08}_{-0.07}$	11000	$14.1^{+0.2}_{-0.3}$
GW190915-235702	$59.9^{+7.5}_{-6.4}$	$25.3^{+3.2}_{-2.7}$	$35.3^{+9.5}_{-6.4}$	$24.4^{+5.6}_{-6.1}$	$0.02^{+0.20}_{-0.25}$	$1.62^{+0.71}_{-0.61}$	$0.30^{+0.11}_{-0.10}$	$57.2^{+7.1}_{-6.0}$	$0.70^{+0.09}_{-0.11}$	400	$13.6^{+0.2}_{-0.3}$
GW190924-021846	$13.9^{+5.1}_{-1.0}$	$5.8^{+0.2}_{-0.2}$	$8.9^{+7.0}_{-2.0}$	$5.0^{+1.4}_{-1.9}$	$0.03^{+0.30}_{-0.09}$	$0.57^{+0.22}_{-0.22}$	$0.12^{+0.04}_{-0.04}$	$13.3^{+5.2}_{-1.0}$	$0.67^{+0.05}_{-0.05}$	360	$11.5^{+0.3}_{-0.4}$
GW190929-012149	$104.3^{+34.9}_{-25.2}$	$35.8^{+14.9}_{-8.2}$	$80.8^{+33.0}_{-33.2}$	$24.1^{+19.3}_{-10.6}$	$0.01^{+0.34}_{-0.33}$	$2.13^{+3.65}_{-1.05}$	$0.38^{+0.49}_{-0.17}$	$101.5^{+33.6}_{-25.3}$	$0.66^{+0.20}_{-0.31}$	2200	$10.1^{+0.6}_{-0.8}$
GW190930-133541	$20.3^{+8.9}_{-1.5}$	$8.5^{+0.5}_{-0.5}$	$12.3^{+12.4}_{-2.3}$	$7.8^{+1.7}_{-3.3}$	$0.14^{+0.31}_{-0.15}$	$0.76^{+0.36}_{-0.32}$	$0.15^{+0.06}_{-0.06}$	$19.4^{+9.2}_{-1.5}$	$0.72^{+0.07}_{-0.06}$	1700	$9.5^{+0.3}_{-0.5}$

TABLE VI. Median and 90% symmetric credible intervals on selected source parameters. The columns show source total mass M , chirp mass $\mathcal{M}$ and component masses m_i , dimensionless effective inspiral spin χ_{eff} , luminosity distance D_L , redshift z , final mass M_f , final spin χ_f , and sky localization $\Delta\Omega$. The sky localization is the area of the 90% credible region. For GW190425 we show the results using the high-spin prior ($|\vec{\chi}_i| \leq 0.89$). We also report the network matched filter SNR for all events. These SNRs are from LALInference IMRPhenomPv2 runs since RIFT does not produce the SNRs automatically, except for GW190425 and GW190426-152155 which use the SNRs from fiducial runs, and GW190412, GW190521, and GW190814, which use IMRPhenomPv3HM SNRs. For GW190521 we report results averaged over three waveform families, in contrast to the results highlighting one waveform family in [34].